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23BTRCL201

CSE-AIML

HCLTFP2316251

SECTION A: ADVANCED SQL

Question 1: Database Design and Advanced Queries (8 Marks)

a. Display employee names, department names, and manager names using joins. (2 marks

```
49
50 •  SELECT e.EmployeeName,
51           d.DepartmentName,
52           m.EmployeeName AS ManagerName
53    FROM Employees e
54    LEFT JOIN Departments d
55        ON e.DepartmentID = d.DepartmentID
56    LEFT JOIN Employees m
57        ON e.ManagerID = m.EmployeeID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EmployeeName	DepartmentName	ManagerName
Arun	IT	NULL
Bharath	IT	Arun
Charan	IT	Arun
Deepak	IT	Bharath
Eshwar	HR	NULL
Farhan	HR	Eshwar
Ganesh	HR	Eshwar
Harish	Finance	NULL
Ishan	Finance	Harish
Jay	Finance	Harish
Kiran	Marketing	NULL

Result 1

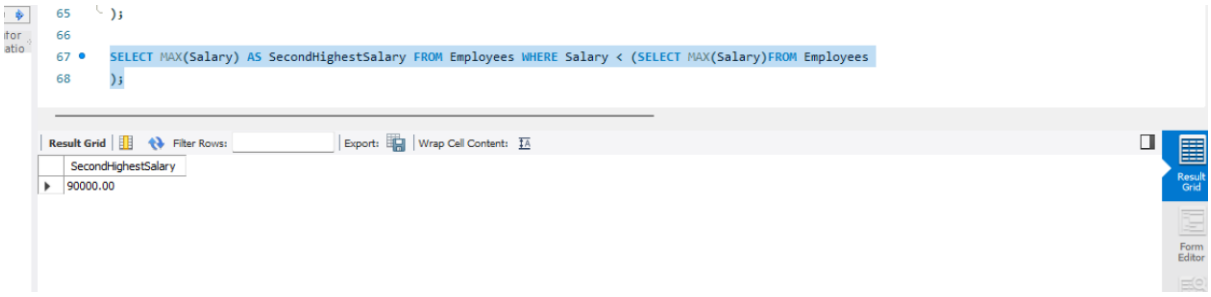
b. Find employees earning more than their department's average salary. (2 marks)

```
58
59  SELECT EmployeeName, DepartmentID, Salary
60  FROM Employees e
61  WHERE Salary > (
62      SELECT AVG(Salary)
63      FROM Employees
64      WHERE DepartmentID = e.DepartmentID
65  );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EmployeeName	DepartmentID	Salary
Arun	1	90000.00
Bharath	1	70000.00
Eshwar	2	80000.00
Harish	3	95000.00
Kiran	4	75000.00

c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. (2 marks)



d. Rank employees by salary within each department and display the top three employees per department, retaining ties at the cutoff. (2 marks)

```
70  WITH RankedEmployees AS (
71      SELECT EmployeeID, EmployeeName,DepartmentID, Salary,DENSE_RANK() OVER (
72      PARTITION BY DepartmentID
73                  ORDER BY Salary DESC
74              ) AS SalaryRank
75      FROM Employees
76  )
77  SELECT *
78  FROM RankedEmployees
79  WHERE SalaryRank <= 3
80  ORDER BY DepartmentID, Salary DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EmployeeID	EmployeeName	DepartmentID	Salary	SalaryRank
1	Arun	1	90000.00	1
2	Bharath	1	70000.00	2
3	Charan	1	60000.00	3
4	Deepak	1	60000.00	3
5	Eshwar	2	80000.00	1
6	Farhan	2	50000.00	2
15	Praveen	2	50000.00	2
7	Ganesh	2	45000.00	3
8	Harish	3	95000.00	1
9	Ishan	3	65000.00	2
10	Jay	3	55000.00	3
11	Kiran	4	75000.00	1
12	Lokesh	4	55000.00	2
13	Manoj	4	40000.00	3

Question 2: Create a Sales table with sale id ,cutomer id ,sale date , product name ,quantity ,unitprice

a.Calculate total revenue for each product. (2 marks)

```
107
108  SELECT ProductName, SUM(Quantity * UnitPrice) AS TotalRevenue
109  FROM Sales
110  GROUP BY ProductName;
111
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ProductName	TotalRevenue
Laptop	350000.00
Mouse	14500.00
Keyboard	13500.00
Monitor	72000.00

b. Calculate monthly revenue and month-over-month revenue change using a window function. (2 marks)

```sql
WITH MonthlySales AS (
    SELECT DATE_FORMAT(SaleDate, '%Y-%m') AS SaleMonth,
           SUM(Quantity * UnitPrice) AS Revenue
    FROM Sales
    GROUP BY DATE_FORMAT(SaleDate, '%Y-%m')
),
RevenueChanges AS (
    SELECT SaleMonth,
           Revenue,
           LAG(Revenue) OVER (
               ORDER BY SaleMonth
           ) AS PreviousRevenue
    FROM MonthlySales
)
SELECT SaleMonth, Revenue, PreviousRevenue,
       Revenue - PreviousRevenue AS RevenueChange
FROM RevenueChanges
ORDER BY SaleMonth;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SaleMonth	Revenue	PreviousRevenue	RevenueChange
2025-01	107000.00	NULL	NULL
2025-02	79000.00	107000.00	-28000.00
2025-03	68000.00	79000.00	-11000.00
2025-04	107000.00	68000.00	39000.00
2025-05	89000.00	107000.00	-18000.00

Result 9

c. Rewrite the following query to improve the potential for index usage and explain why your version may perform better. (2 marks)

```sql
ORDER BY SaleMonth;

SELECT *
FROM Employees
WHERE JoiningDate >= '2021-03-15'
  AND JoiningDate < '2022-11-05';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

EmployeeID	EmployeeName	DepartmentID	Salary	JoiningDate	ManagerID
2	Bharath	1	70000.00	2021-03-15	1
3	Charan	1	60000.00	2022-05-20	1
6	Farhan	2	50000.00	2021-08-10	5
12	Lokesh	4	55000.00	2022-01-16	11

Question 3: A company maintains millions of employee and transaction records. Its reporting queries are running slowly

a.An execution plan shows how the database executes a query.

- It identifies table scans, index scans, joins and sorting operations.
- It provides estimated costs and row counts.
- It helps locate slow operations and unnecessary full table scans.
- Developers can improve performance by adding appropriate indexes or rewriting queries.

b. Recommend an appropriate index strategy for queries filtering employees by DepartmentID and Salary. Explain your reasoning. (2 marks)

```
135
136 •     CREATE INDEX idx_employee_dept_salary
137     ON Employees (DepartmentID, Salary);
```

- Creating index strategy for the filtering employees by the department id and the salary I would like to sugget the simplest query to by using the "ON" where it can compare the employees by comparing the salary and the departmentID.

c. Write a query to identify duplicate customer email addresses and display only addresses occurring more than once. (2 marks)

- I would like to recommend the query which identify the duplicate customer email address and display the address occurring only once
- I'll be using the count and having where it is used to filter the email address for
- And ill use GROUP BY to comparing and filtering the email address from the customers table and the values

  example:

  SELECT Email, COUNT(*) FROM Customers GROUP BY Email HAVING COUNT(*) > 1;

SECTION B: LINUX

Question 4: Log Analysis (5 Marks)

- FIRST create the file and inside the file add the data and error lines if the file is not created already

- After creating the file

a. Find all lines containing ERROR. (1 mark)

- grep 'ERROR' "$LOG" using this command helps to find the error lines in the file which is /var/log/application.log

```
[ec2-user@ip-172-31-45-35 ~]$ grep 'ERROR' "$LOG"
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR database-service Connection failed
2026-10-28 11:30:45 ERROR payment-service Timeout occurred
2026-10-29 09:15:00 ERROR auth-service Login failed
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR database-service Connection failed
```

b. Display the 20 most recent error entries in the file. (1 mark)

- grep 'ERROR' "$LOG" | tail -n 20 Tail is used to display the last lines of the file here I jave inserted file few error log file show it is displaying how many logs ive insterted it is showing them

```
[ec2-user@ip-172-31-45-35 ~]$ grep 'ERROR' "$LOG" | tail -n 20
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR database-service Connection failed
2026-10-28 11:30:45 ERROR payment-service Timeout occurred
2026-10-29 09:15:00 ERROR auth-service Login failed
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR database-service Connection failed
2026-10-28 11:30:45 ERROR payment-service Timeout occurred
2026-10-29 09:15:00 ERROR auth-service Login failed
```

c. Count the total number of error entries. (1 mark)

- grep -c 'ERROR' "$LOG"

```
[ec2-user@ip-172-31-45-35 ~]$ grep -c 'ERROR' "$LOG"
8
```

d. Extract error entries for a specified date into a separate file. (1 mark)

- grep '^2026-10-28 .*ERROR' "$LOG" > ~/errors_2026-10-28.log cat ~/errors_2026-10-28.log we are saving the file bbythe date of 2026-10-28 and the cat is used to display the saved error log files ive saved it twice it is showing 8

```
[ec2-user@ip-172-31-45-35 ~]$ grep '^2026-10-28 .*ERROR' "$LOG" > ~/errors_2026-10-28.log
cat ~/errors_2026-10-28.log
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR database-service Connection failed
2026-10-28 11:30:45 ERROR payment-service Timeout occurred
2026-10-28 10:15:23 ERROR payment-service Transaction failed
2026-10-28 11:05:12 ERROR database-service Connection failed
2026-10-28 11:30:45 ERROR payment-service Timeout occurred
```

e.Explain how you would calculate the number of errors occurring in each hour. (1 mark)

- awk filters error entries, extracts the hour from the timestamp, and counts errors per hour. sort displays the results in hour order.

Question 5: Permissions and System Troubleshooting

a. Create a group named finance and a report file, assuming you have the required privileges. (1 mark)

- sudo groupadd finance sudo touch report.txt using these command to creating the the report finance in the group finance and sudo is used to execute the command

```
[ec2-user@ip-172-31-45-35 ~]$ sudo groupadd finance
sudo touch report.txtsudo groupadd finance
sudo touch report.txt
[ec2-user@ip-172-31-45-35 ~]$ sudo groupadd finance
sudo touch report.txt
```

b. Set permissions so the owner can read and write, the finance group can read, and other users have no access. (1 mark)

```
[ec2-user@ip-172-31-45-35 ~]$ sudo chmod 640 report.txt
[ec2-user@ip-172-31-45-35 ~]$ ls -l report.txt
-rw-r-----. 1 ec2-user finance 0 Oct 28 06:23 report.txt
```

c.Identify the processes consuming the most CPU. (1 mark)

- using top command

```
top - 06:32:05 up 13 days,  1:58,  1 user,  load average: 0.00, 0.00, 0.00
Tasks: 106 total,   1 running, 105 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 98.8 id,  0.0 wa,  0.0 hi,  0.0 si,  1.2 st
MiB Mem :    913.1 total,    200.4 free,    198.5 used,    514.2 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    575.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  174344  18276  11236 S   0.0   2.0   1:11.52 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.23 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:02.16 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:03.86 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:18.20 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.44 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:04.65 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     21 root      rt   0       0      0      0 S   0.0   0.0   0:04.63 migration/1
     22 root      20   0       0      0      0 S   0.0   0.0   0:03.65 ksoftirqd/1
     24 root       0 -20       0      0      0 I   0.0   0.0   0:02.43 kworker/1:0H-kblockd
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     28 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     29 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     30 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     31 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     32 root      20   0       0      0      0 S   0.0   0.0   0:00.61 kauditd
     34 root      20   0       0      0      0 S   0.0   0.0   0:00.38 khungtaskd
     35 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
```

d. Display memory usage and filesystem disk utilization. (1 mark)

- Df -h is used to filesystem utiliztion

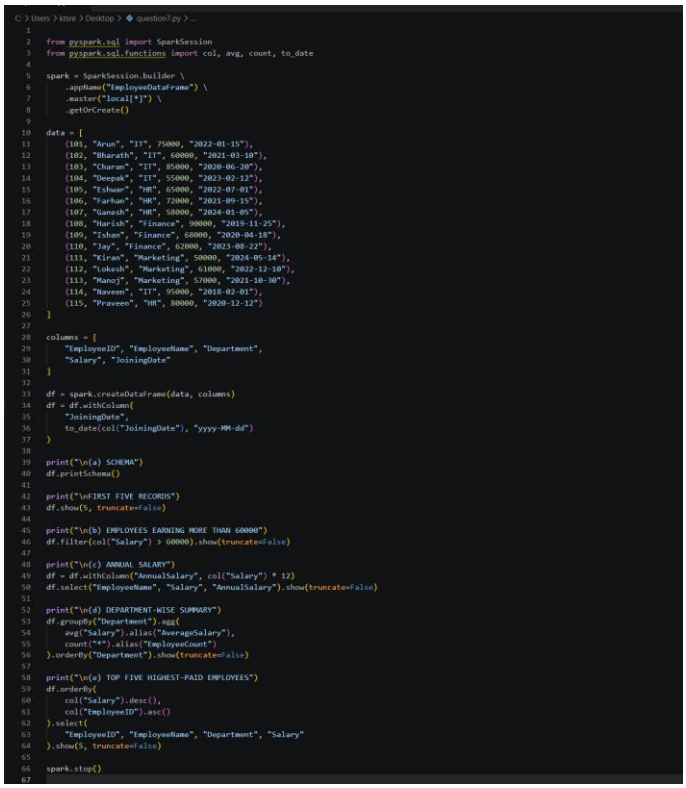
```
[ec2-user@ip-172-31-45-35 ~]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             457M     0  457M   0% /dev/shm
tmpfs             183M  448K  183M   1% /run
efivarfs          128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1    8.0G  1.8G  6.2G  23% /
tmpfs             457M     0  457M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
```

- Free -h used to memoeryusage

```
[ec2-user@ip-172-31-45-35 ~]$ free -h
               total        used        free      shared  buff/cache   available
Mem:           913Mi       197Mi       201Mi       0.0Ki       514Mi       577Mi
Swap:             0B          0B          0B
```

SECTION C: PYSPARK DATA ENGINEERING

Create a PySpark DataFrame



```python
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, avg, count, to_date

spark = SparkSession.builder \
    .appName("EmployeeDataFrame") \
    .master("local[*]") \
    .getOrCreate()

data = [
    (101, "Arun", "IT", 75000, "2022-01-15"),
    (102, "Bharath", "IT", 60000, "2021-03-10"),
    (103, "Charan", "IT", 85000, "2020-06-20"),
    (104, "Deepak", "IT", 55000, "2023-02-12"),
    (105, "Eshwar", "HR", 65000, "2022-07-01"),
    (106, "Farhan", "HR", 72000, "2021-09-15"),
    (107, "Ganesh", "HR", 58000, "2024-01-05"),
    (108, "Hariah", "Finance", 90000, "2019-11-25"),
    (109, "Ishan", "Finance", 68000, "2020-04-18"),
    (110, "Jay", "Finance", 62000, "2023-08-22"),
    (111, "Kiran", "Marketing", 58000, "2024-05-14"),
    (112, "Lokesh", "Marketing", 61000, "2022-12-10"),
    (113, "Manoj", "Marketing", 57000, "2021-10-30"),
    (114, "Naveen", "IT", 95000, "2018-02-01"),
    (115, "Praveen", "HR", 80000, "2020-12-12")
]

columns = [
    "EmployeeID", "EmployeeName", "Department",
    "Salary", "JoiningDate"
]

df = spark.createDataFrame(data, columns)
df = df.withColumn(
    "JoiningDate",
    to_date(col("JoiningDate"), "yyyy-MM-dd")
)

print("\n(a) SCHEMA")
df.printSchema()

print("\nFIRST FIVE RECORDS")
df.show(5, truncate=False)

print("\n(b) EMPLOYEES EARNING MORE THAN 60000")
df.filter(col("Salary") > 60000).show(truncate=False)

print("\n(c) ANNUAL SALARY")
df = df.withColumn("AnnualSalary", col("Salary") * 12)
df.select("EmployeeName", "Salary", "AnnualSalary").show(truncate=False)

print("\n(d) DEPARTMENT-WISE SUMMARY")
df.groupBy("Department").agg(
    avg("Salary").alias("AverageSalary"),
    count("*").alias("EmployeeCount")
).orderBy("Department").show(truncate=False)

print("\n(e) TOP FIVE HIGHEST-PAID EMPLOYEES")
df.orderBy(
    col("Salary").desc(),
    col("EmployeeID").asc()
).select(
    "EmployeeID", "EmployeeName", "Department", "Salary"
).show(5, truncate=False)

spark.stop()
```

Question 7 PySpark DataFrame Operations:

a. **Display the schema and first five records. (1 mark)**

```
(a) SCHEMA
root
 |-- EmployeeID: long (nullable = true)
 |-- EmployeeName: string (nullable = true)
 |-- Department: string (nullable = true)
 |-- Salary: long (nullable = true)
 |-- JoiningDate: date (nullable = true)
```

b. Filter employees whose monthly salary is greater than ₹60,000. (1 mark)

```
(b) EMPLOYEES EARNING MORE THAN 60000
+----------+------------+----------+------+-----------+
|EmployeeID|EmployeeName|Department|Salary|JoiningDate|
+----------+------------+----------+------+-----------+
|101       |Arun        |IT        |75000 |2022-01-15 |
|103       |Charan      |IT        |85000 |2020-06-20 |
|105       |Eshwar      |HR        |65000 |2022-07-01 |
|106       |Farhan      |HR        |72000 |2021-09-15 |
|108       |Harish      |Finance   |90000 |2019-11-25 |
|109       |Ishan       |Finance   |68000 |2020-04-18 |
|110       |Jay         |Finance   |62000 |2023-08-22 |
|112       |Lokesh      |Marketing |61000 |2022-12-10 |
|114       |Naveen      |IT        |95000 |2018-02-01 |
|115       |Praveen     |HR        |80000 |2020-12-12 |
+----------+------------+----------+------+-----------+
```

c. Add an AnnualSalary column calculated from the monthly salary. (1 mark)

```
(c) ANNUAL SALARY
+------------+------+------------+
|EmployeeName|Salary|AnnualSalary|
+------------+------+------------+
|Arun        |75000 |900000      |
|Bharath     |60000 |720000      |
|Charan      |85000 |1020000     |
|Deepak      |55000 |660000      |
|Eshwar      |65000 |780000      |
|Farhan      |72000 |864000      |
|Ganesh      |58000 |696000      |
|Harish      |90000 |1080000     |
|Ishan       |68000 |816000      |
|Jay         |62000 |744000      |
|Kiran       |50000 |600000      |
|Lokesh      |61000 |732000      |
|Manoj       |57000 |684000      |
|Naveen      |95000 |1140000     |
|Praveen     |80000 |960000      |
+------------+------+------------+
```

d. Group employees by department and calculate the average salary and employee count. (2 marks)

```
(d) DEPARTMENT-WISE SUMMARY
+----------+-----------------+-------------+
|Department|AverageSalary    |EmployeeCount|
+----------+-----------------+-------------+
|Finance   |73333.33333333333|3            |
|HR        |68750.0          |4            |
|IT        |74000.0          |5            |
|Marketing |56000.0          |3            |
+----------+-----------------+-------------+
```

Question 8: Data Cleaning and Aggregation

a. **Read the CSV file using an appropriate schema and header configuration. (1 mark)**

```
(a) ORIGINAL TRANSACTIONS
+-------------+----------+---------------+--------+--------+---------+
|TransactionID|CustomerID|TransactionDate|Product |Quantity|UnitPrice|
+-------------+----------+---------------+--------+--------+---------+
|T001         |C101      |2026-10-01     |Laptop  |1       |55000.0  |
|T002         |C102      |02/10/2026     |Mouse   |2       |500.0    |
|T003         |C103      |2026-10-03     |Keyboard|1       |1500.0   |
|T004         |C104      |2026-10-04     |Laptop  |1       |55000.0  |
|T002         |C102      |02/10/2026     |Mouse   |2       |500.0    |
|T005         |NULL      |2026-10-05     |Mouse   |3       |500.0    |
|T006         |C106      |invalid-date   |Keyboard|2       |1500.0   |
|T007         |C107      |2026-10-07     |NULL    |1       |2000.0   |
|T008         |C108      |2026-10-08     |Monitor |NULL    |12000.0  |
|T009         |C109      |2026-10-09     |Monitor |2       |NULL     |
+-------------+----------+---------------+--------+--------+---------+
```

b. **Identify and remove duplicate transactions using a suitable business key. (1 mark)**

```
(b) AFTER REMOVING DUPLICATES
+-------------+----------+---------------+--------+--------+---------+
|TransactionID|CustomerID|TransactionDate|Product |Quantity|UnitPrice|
+-------------+----------+---------------+--------+--------+---------+
|T001         |C101      |2026-10-01     |Laptop  |1       |55000.0  |
|T002         |C102      |02/10/2026     |Mouse   |2       |500.0    |
|T003         |C103      |2026-10-03     |Keyboard|1       |1500.0   |
|T004         |C104      |2026-10-04     |Laptop  |1       |55000.0  |
|T005         |NULL      |2026-10-05     |Mouse   |3       |500.0    |
|T006         |C106      |invalid-date   |Keyboard|2       |1500.0   |
|T007         |C107      |2026-10-07     |NULL    |1       |2000.0   |
|T008         |C108      |2026-10-08     |Monitor |NULL    |12000.0  |
|T009         |C109      |2026-10-09     |Monitor |2       |NULL     |
+-------------+----------+---------------+--------+--------+---------+
```

c. **Handle missing values in required fields and explain your chosen approach. (2 marks)**

- the dropna() function removes records containing missing values in the specified required columns.
- This approach is chosen because missing transaction details can lead to incorrect revenue calculations and unreliable analysis.

d.Convert TransactionDate into a date type and handle invalid dates. (1 mark)

```
+-------------+----------+--------+--------+---------+---------------+
|TransactionID|CustomerID|Product |Quantity|UnitPrice|TransactionDate|
+-------------+----------+--------+--------+---------+---------------+
|T001         |C101      |Laptop  |1       |55000.0  |2026-10-01     |
|T002         |C102      |Mouse   |2       |500.0    |2026-10-02     |
|T003         |C103      |Keyboard|1       |1500.0   |2026-10-03     |
|T004         |C104      |Laptop  |1       |55000.0  |2026-10-04     |
+-------------+----------+--------+--------+---------+---------------+
```

e. Calculate revenue using Quantity × UnitPrice. (1 mark)

```
(e) TRANSACTIONS WITH REVENUE
+-------------+----------+--------+--------+---------+---------------+-------+
|TransactionID|CustomerID|Product |Quantity|UnitPrice|TransactionDate|Revenue|
+-------------+----------+--------+--------+---------+---------------+-------+
|T001         |C101      |Laptop  |1       |55000.0  |2026-10-01     |55000.0|
|T002         |C102      |Mouse   |2       |500.0    |2026-10-02     |1000.0 |
|T003         |C103      |Keyboard|1       |1500.0   |2026-10-03     |1500.0 |
|T004         |C104      |Laptop  |1       |55000.0  |2026-10-04     |55000.0|
+-------------+----------+--------+--------+---------+---------------+-------+
```

f.Calculate total revenue by product and display the results in descending order. (1 mark)

```
(f) TOTAL REVENUE BY PRODUCT
+--------+------------+
|Product |TotalRevenue|
+--------+------------+
|Laptop  |110000.0    |
|Keyboard|1500.0      |
|Mouse   |1000.0      |
+--------+------------+
```

Code:

```python
print("\n(a) Original data")
df.show(truncate=False)

# (b) Remove repeated transactions
df = df.dropDuplicates(["TransactionID"])

print("\n(b) After removing duplicates")
df.show(truncate=False)

# (c) Remove rows with missing details
df = df.dropna(subset=[
    "TransactionID", "CustomerID", "TransactionDate",
    "Product", "Quantity", "UnitPrice"
])

# Keep only valid quantities and prices
df = df.filter((col("Quantity") > 0) & (col("UnitPrice") > 0))

print("\n(c) After removing missing values")
df.show(truncate=False)

# (d) Convert the dates into a proper date format
df = df.withColumn(
    "NewDate",
    coalesce(
        to_date(try_to_timestamp(col("TransactionDate"), lit("yyyy-MM-dd"))),
        to_date(try_to_timestamp(col("TransactionDate"), lit("dd/MM/yyyy"))),
        to_date(try_to_timestamp(col("TransactionDate"), lit("MM/dd/yyyy")))
    )
)

# Remove rows where the date is invalid
df = df.filter(col("NewDate").isNotNull())

df = df.drop("TransactionDate")
df = df.withColumnRenamed("NewDate", "TransactionDate")

print("\n(d) After cleaning dates")
df.show(truncate=False)

# (e) Find the revenue for each transaction
df = df.withColumn("Revenue", col("Quantity") * col("UnitPrice"))

print("\n(e) Revenue for each transaction")
df.show(truncate=False)

# (f) Find total revenue for each product
print("\n(f) Total revenue by product")

df.groupBy("Product").agg(
    total("Revenue").alias("TotalRevenue")
).orderBy(col("TotalRevenue").desc()).show(truncate=False)

# Close Spark
spark.stop()
```

Question 9: PySpark Joins and Performance

a. Load both datasets into DataFrames. (1 mark)

```
Customers Data
+----------+------------+---------+
|CustomerID|CustomerName|     City|
+----------+------------+---------+
|      C101|       Rahul|Bengaluru|
|      C102|       Priya|  Chennai|
|      C103|        Arun|Hyderabad|
|      C104|       Meena|Bengaluru|
|      C105|       Kiran|   Mysuru|
+----------+------------+---------+
```

b. Join the datasets to display orders with customer details. (2 marks)

```
Orders with Customer Details
+----------+------------+---------+-------+-----------+----------+
|CustomerID|CustomerName|City     |OrderID|OrderAmount|OrderDate |
+----------+------------+---------+-------+-----------+----------+
|C101      |Rahul       |Bengaluru|O101   |5000       |2026-10-01|
|C102      |Priya       |Chennai  |O102   |3000       |2026-10-02|
|C101      |Rahul       |Bengaluru|O103   |2000       |2026-10-03|
|C104      |Meena       |Bengaluru|O104   |4500       |2026-10-04|
+----------+------------+---------+-------+-----------+----------+
```

c.Identify customers who have never placed an order. (1 mark)

```
Customers Without Orders
+----------+------------+---------+
|CustomerID|CustomerName|City     |
+----------+------------+---------+
|C103      |Arun        |Hyderabad|
|C105      |Kiran       |Mysuru   |
+----------+------------+---------+
```

d. Calculate the total order value by city. (1 mark)

```
Total Order Value by City
+---------+---------------+
|City     |TotalOrderValue|
+---------+---------------+
|Chennai  |3000           |
|Bengaluru|11500          |
+---------+---------------+
```

e. **Explain when a broadcast join is useful and what conditions make it appropriate. (1 mark)**

- A broadcast join is useful when one DataFrame is small enough to fit in the memory of each worker. Spark sends the smaller DataFrame to all workers, reducing data shuffling and improving performance.

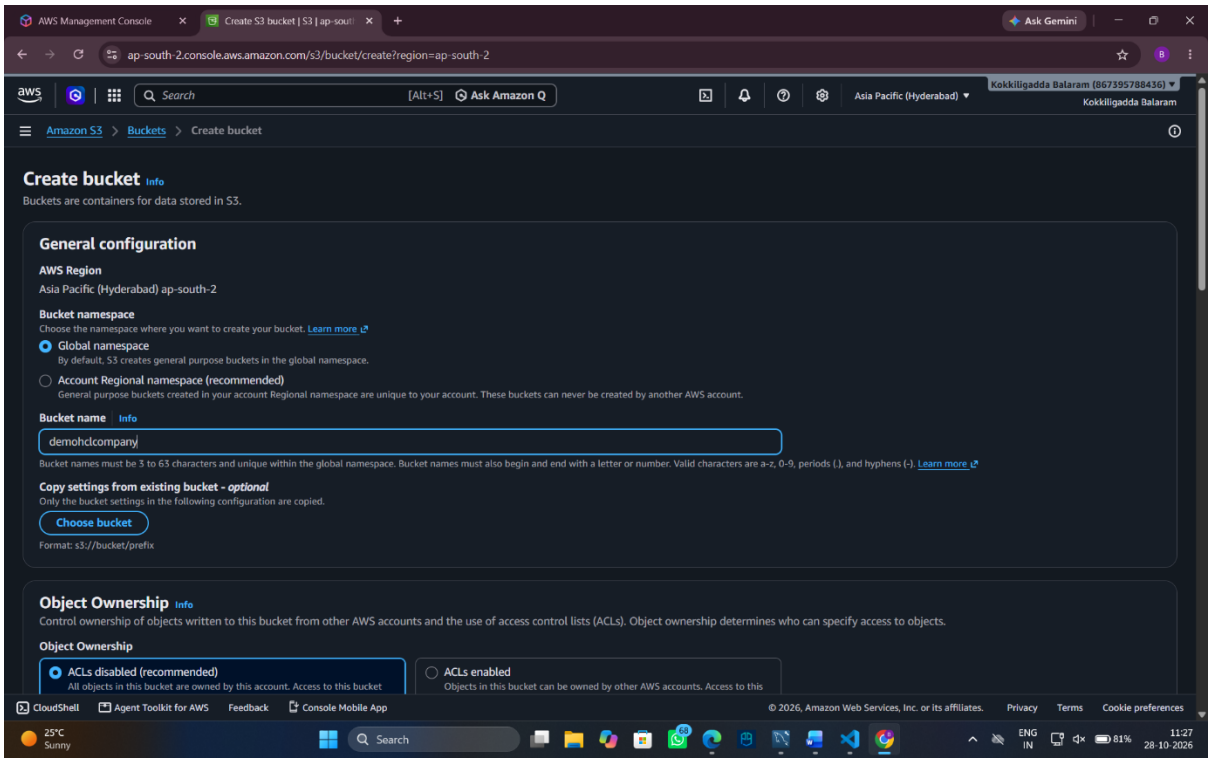
f. **f. Explain the difference between transformations and actions, giving one example of each. (1 mark)**

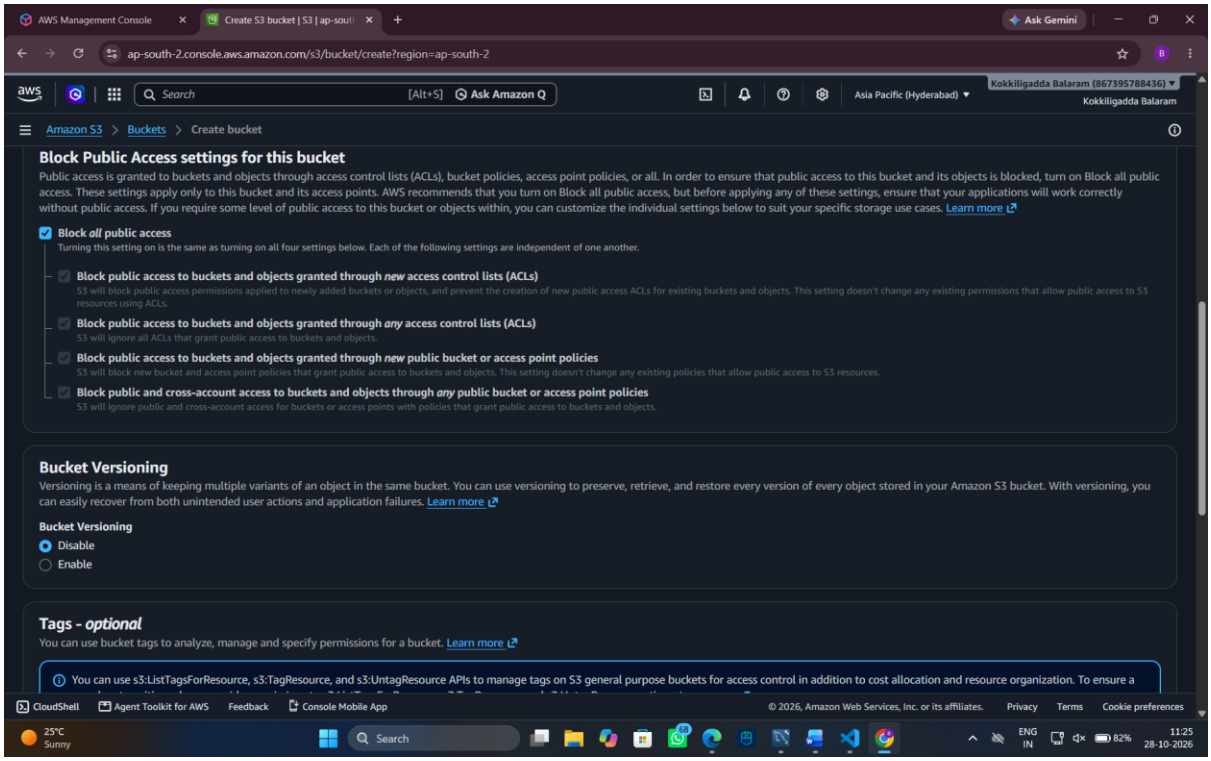
- A broadcast join is useful when one DataFrame is small enough to fit in the memory of each worker. Spark sends the smaller DataFrame to all workers, reducing data shuffling and improving performance.

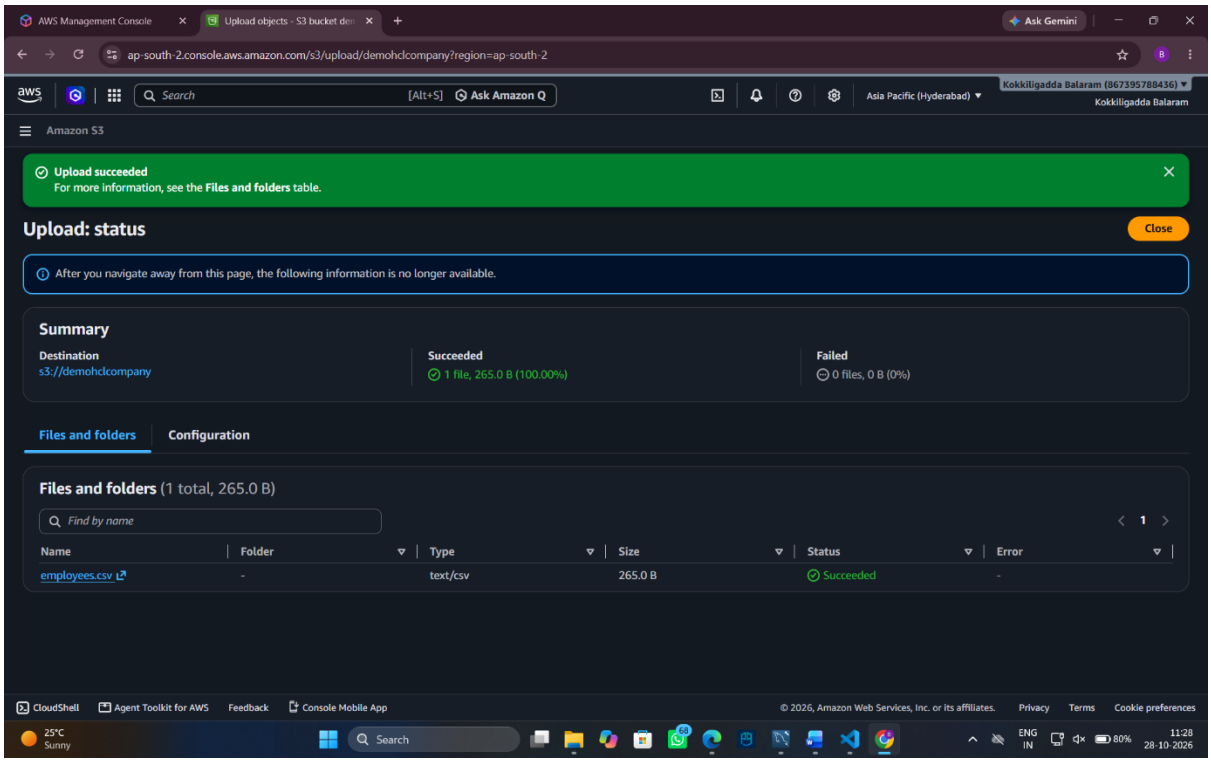
SECTION D: AWS CLOUD SERVICES:

Question 10: S3 and IAM Security:

a. Create an S3 bucket with Block Public Access enabled and upload a sample file. (1 mark)







c.Explain how you would verify that an unauthorized user cannot read the file.

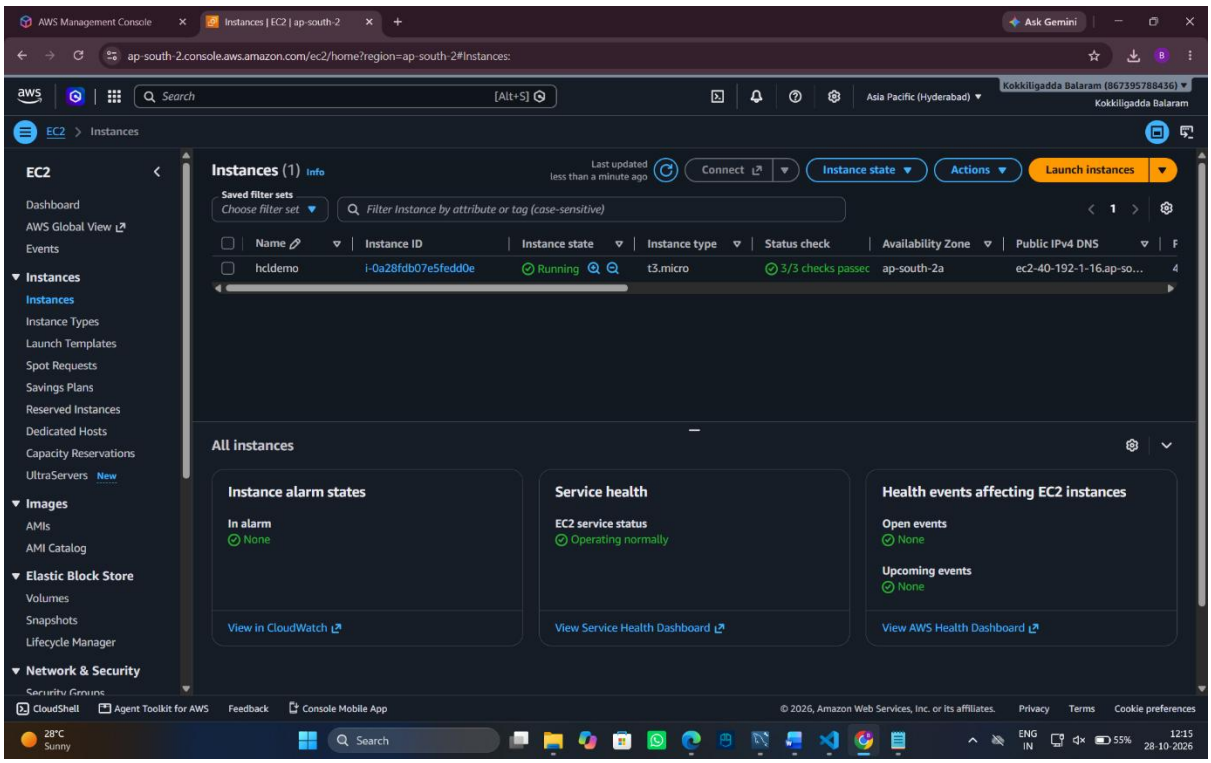
- Sign in using an IAM test user that does not have permission to access the bucket.
- Attempt to open or download employee_report.txt.
- Verify that AWS returns Access Denied.

d.Explain how encryption at rest protects the stored data.

- Encryption at rest protects stored S3 data by converting it into an unreadable encrypted form.
- Amazon S3 automatically encrypts all stored data at rest using SSE-S3 (AES-256) by default at no extra cost.

Question 11: EC2 Troubleshooting and CloudWatch (5 Marks):

a. Check the EC2 instance state and status checks. (1 mark):



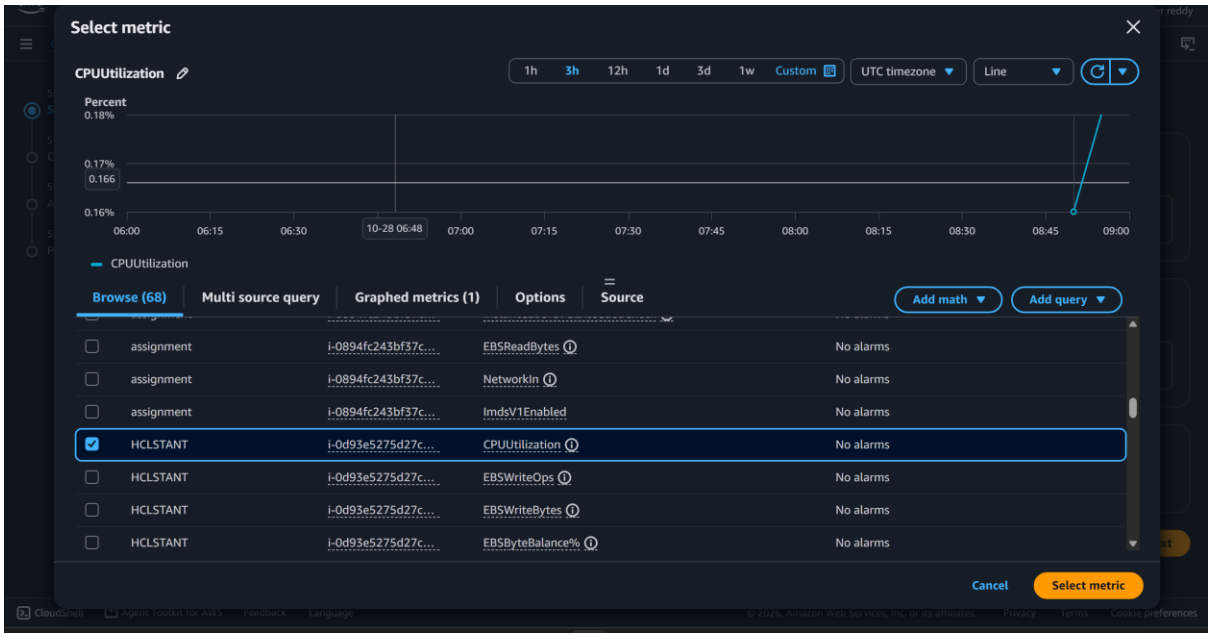
b. Explain how security groups and route tables can affect application connectivity. (1 mark):

- Security groups: Allow or block traffic through specified ports. HTTP uses port 80 and HTTPS uses port 443.

- • Route tables: Direct network traffic. Incorrect routes or a missing internet gateway route can prevent users from reaching the application.

c. Explain how to configure a CloudWatch alarm when CPU utilization exceeds 85% for a defined evaluation period. (2 marks)

- Open Amazon CloudWatch and select Alarms and Create alarm.
- Select the EC2 instance's CPUUtilization metric.
- Set the CPU utilization greater than 85% and choose an evaluation period of 5 minutes.
- Name the alarm and create it.
- The alaram will get on when it come to the threshold 85% .





d. Identify one method for investigating the root cause of high CPU utilization. (1 mark)

- We'll use the command "top" to investigate the root cause of high cpu utilization it should be done by connecting the ec2 instance to the server.

Question 12: Event-Driven Processing (5 Marks):

(a) Identify a suitable service for executing the processing code. (1 mark)

- AWS Lambda is suitable for running processing code automatically without managing servers.

(b) Explain how an S3 upload can trigger processing. (1 mark)

- Configure an S3 Event Notification to trigger a Lambda function whenever a new file is uploaded to the input bucket.

(c) Explain the IAM permissions needed to read the input file and write the results. (1 mark)

- Give the Lambda execution role permission to read input files using
- s3:GetObject and write results using
- s3:PutObject on the required S3 bucket paths only.

(d) Explain how errors and processing logs can be captured. (1 mark)

- Use Amazon CloudWatch Logs to store Lambda execution logs and error messages. CloudWatch can also monitor failures and help troubleshoot problems.

(e) Explain how to prevent duplicate processing or recursive triggering. (1 mark)

- Use a unique file key or event ID to detect duplicate processing.
- Store output files in a separate S3 bucket or a different folder from the input files to prevent the Lambda function from triggering itself repeatedly.

SECTION E: AWS STEP FUNCTIONS:

Question 13: State Machine Design (7 Marks):

a. Identify suitable states for validation, processing, storage, and notification.

The workflow uses AWS Step Functions to coordinate Lambda functions.
Start
Validate File — Task
Is Valid? — Choice
Valid
Transform Data
Calculate Sales
Store in S3
Send Notification
Success
Invalid
Fail State

b. **Explain how a Choice state can route invalid files to a failure path and valid files to processing.**
   A Choice state evaluates a condition in the input data and selects the appropriate transition. For example, if $.isValid is true, the workflow continues to TransformData; otherwise, it moves to FailInvalidFile.

c. **Explain how Retry and Catch can handle transient failures and permanent errors.**
   Retry: Automatically reruns a task after a transient error, using a configured delay and maximum number of attempts.
   Catch: Handles an error that remains after retries by transferring execution to an error-handling state or failure path.

d. **Explain how the state machine can invoke Lambda functions or other supported AWS services.**
   A Task state can invoke a Lambda function using its ARN. Step Functions also supports service integrations with services such as Amazon S3, Amazon SNS and AWS Glue.

e. **Explain how execution history helps troubleshoot a failed workflow.**

Execution history records state transitions, inputs, outputs, errors and timestamps. It identify the failed state, investigate the error and determine where processing stopped.

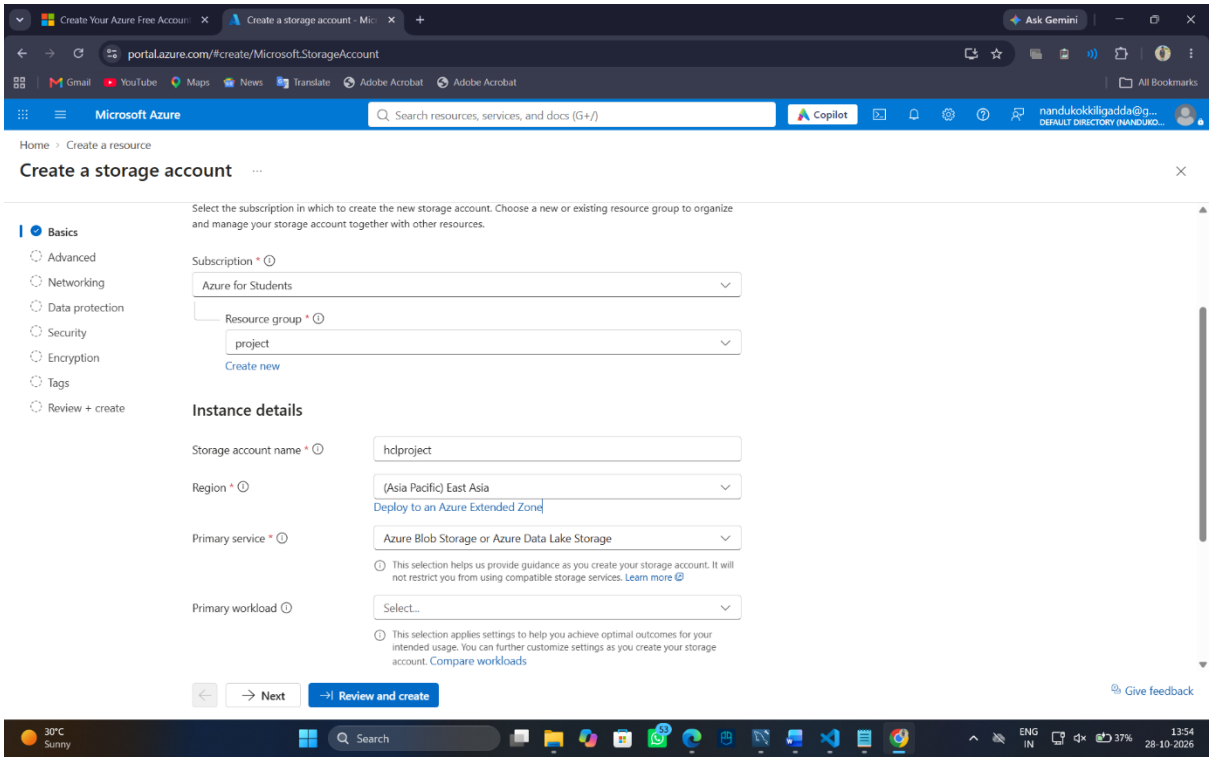
Question 14: Step Functions Implementation

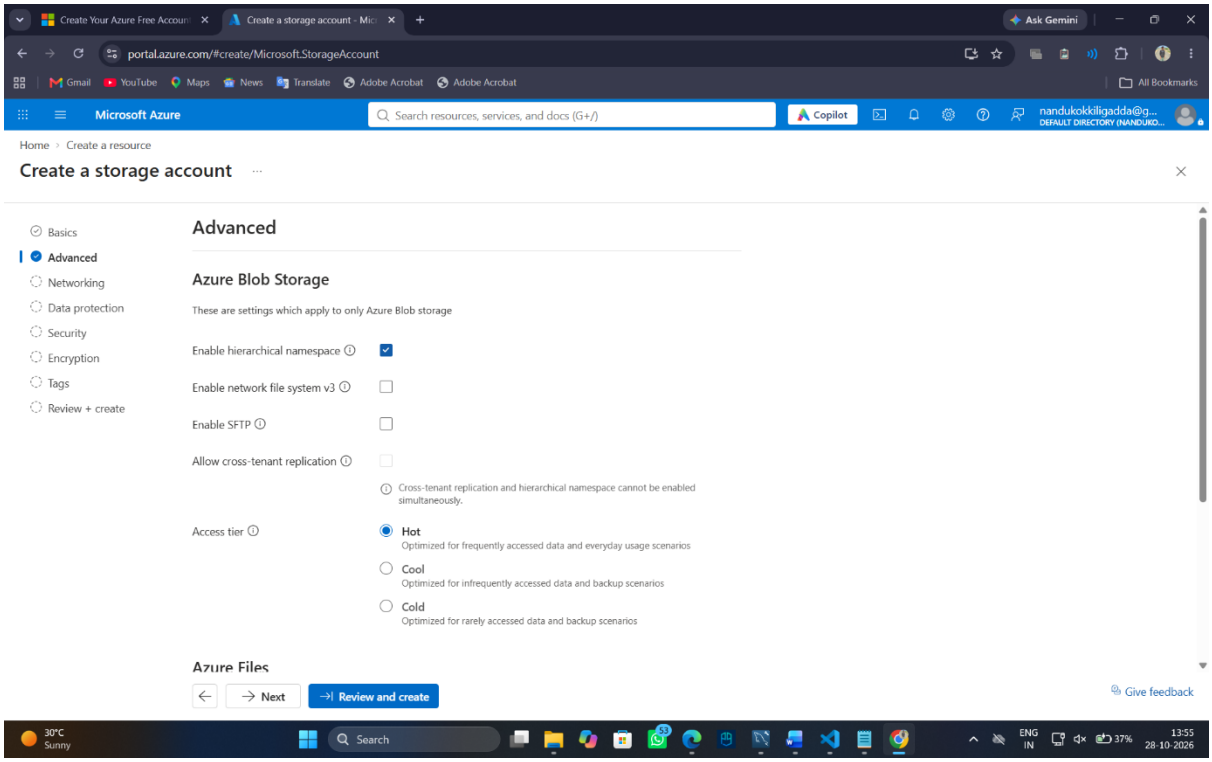
d. **Explain how data from one task is passed to the next task.**
   Step Functions passes the output of one state to the next state as input by default. InputPath, ResultPath and OutputPath can filter or combine input and output. Lambda functions should return JSON-serializable results.
e. **Explain how you would test successful and failed executions.**
   Start an execution with valid input and verify all three tasks succeed.
   Test invalid input and verify validation fails.
   Simulate transient transformation errors to test retries.
   Simulate a loading error and verify the error-handling state is reached.
   Inspect the execution graph and history in the AWS Console.

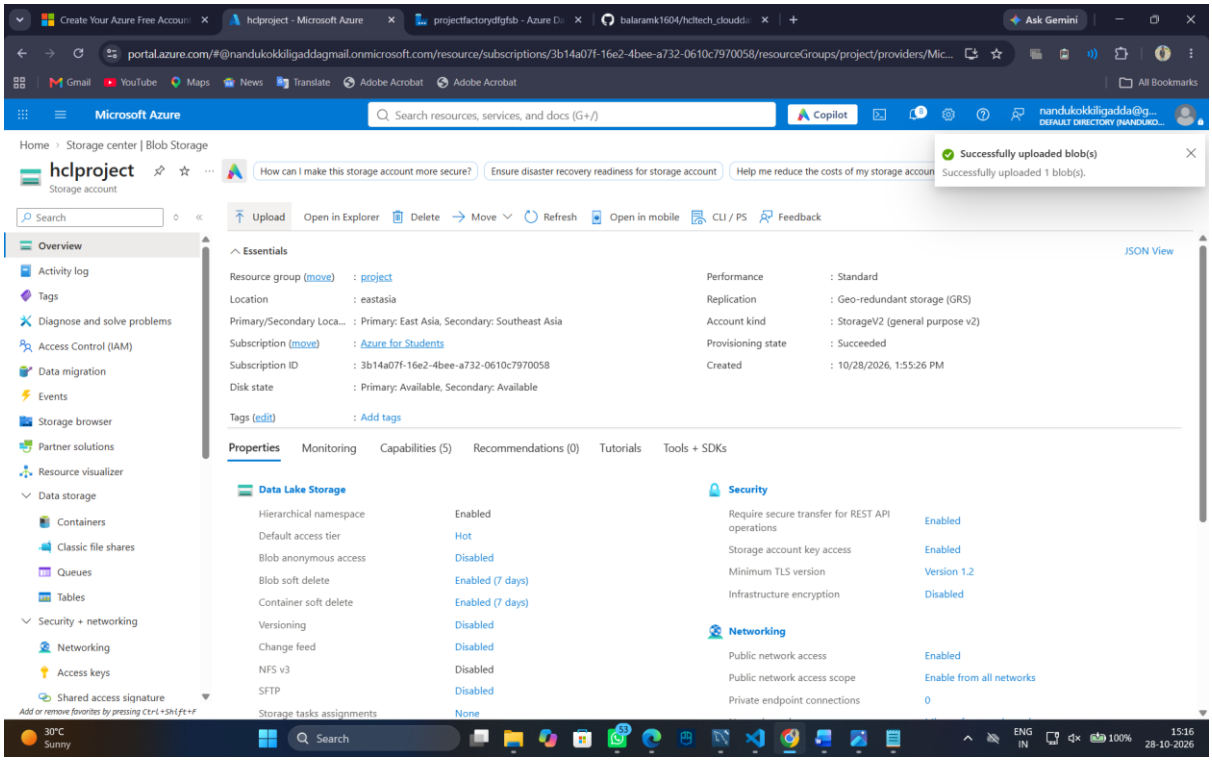
SECTION F: MICROSOFT AZURE

Question 15: Azure Data Lake Storage and Security

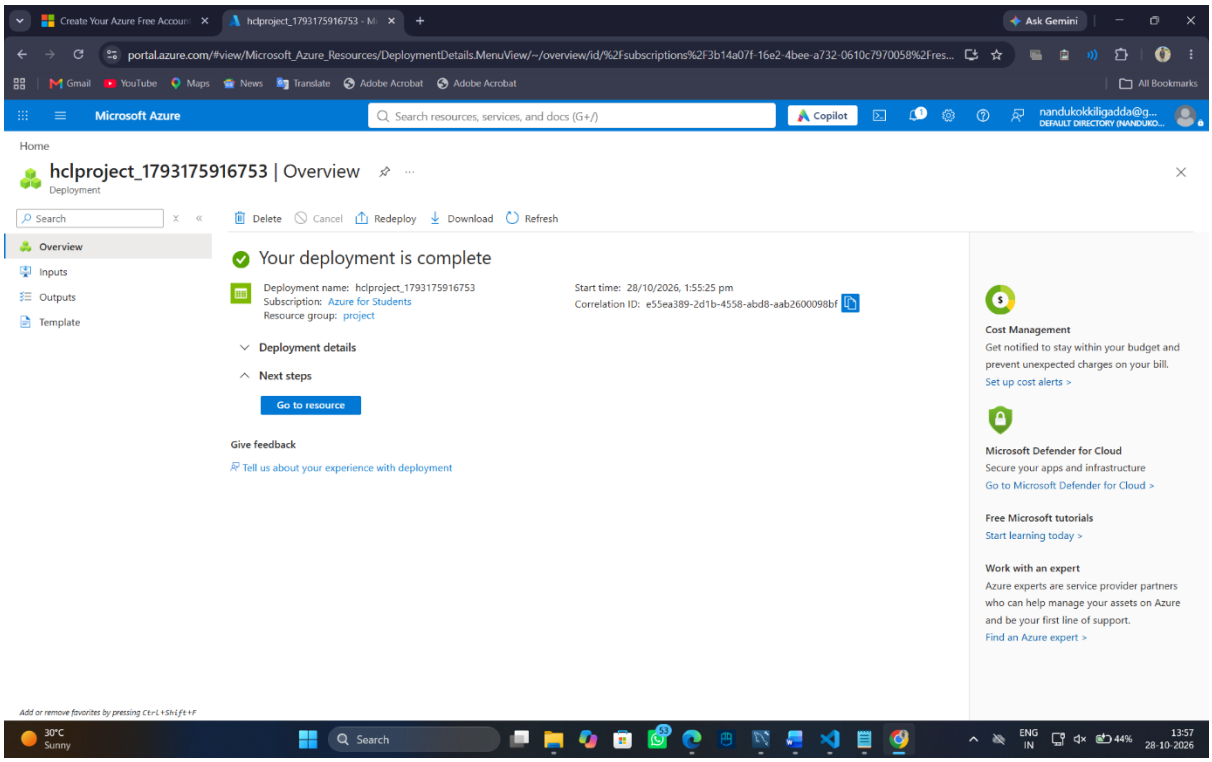
a. **Configure or describe a storage account with hierarchical namespace enabled for Azure Data Lake Storage Gen2.**

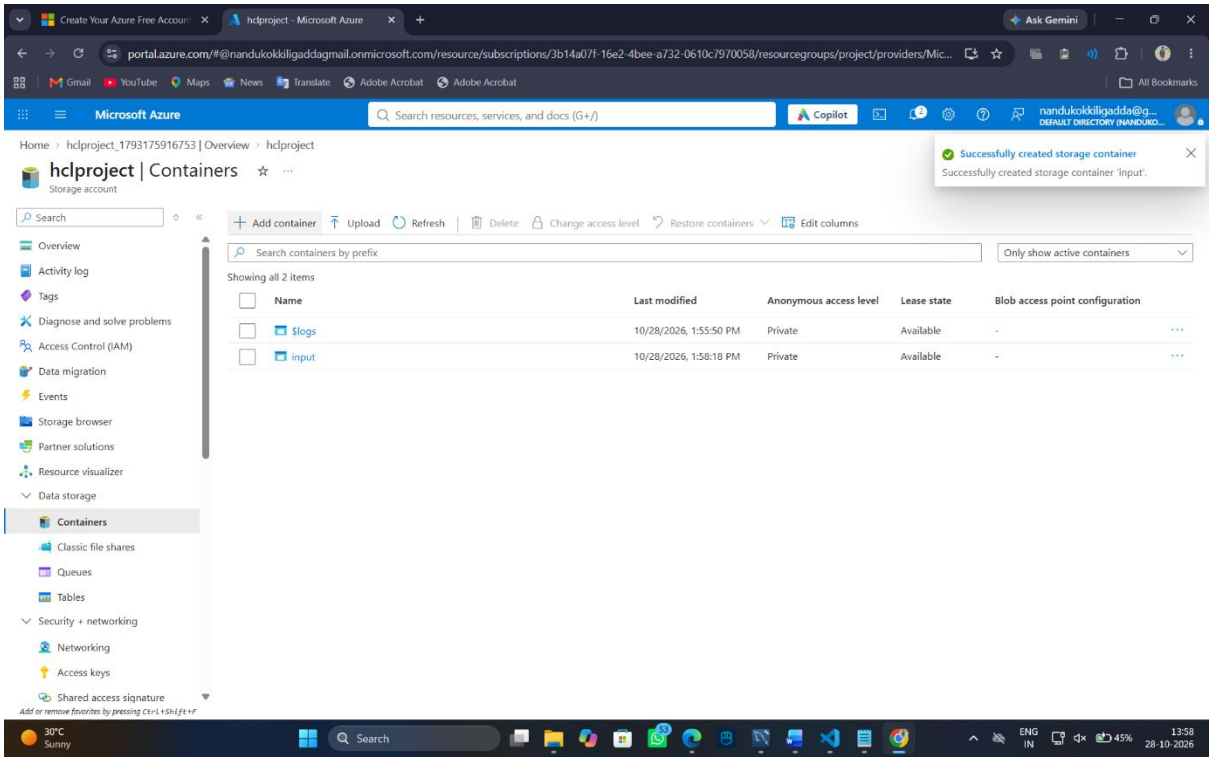

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DEFAULT DIRECTORY (NANDUKO...
Home
Create a resource
Create a storage account
Basics
Advanced
Networking
Data protection
Security
Encryption
Tags
Review + create
Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.
Subscription *
Azure for Students
Resource group *
project
Create new
Instance details
Storage account name *
hclproject
Region *
(Asia Pacific) East Asia
Deploy to an Azure Extended Zone
Primary service *
Azure Blob Storage or Azure Data Lake Storage
This selection helps us provide guidance as you create your storage account. It will not restrict you from using compatible storage services. Learn more
Primary workload
Select...
This selection applies settings to help you achieve optimal outcomes for your intended usage. You can further customize settings as you create your storage account. Compare workloads
Next
Review and create
Give feedback
30°C
Sunny
Search
ENG
IN
37%
13:54
28-10-2026

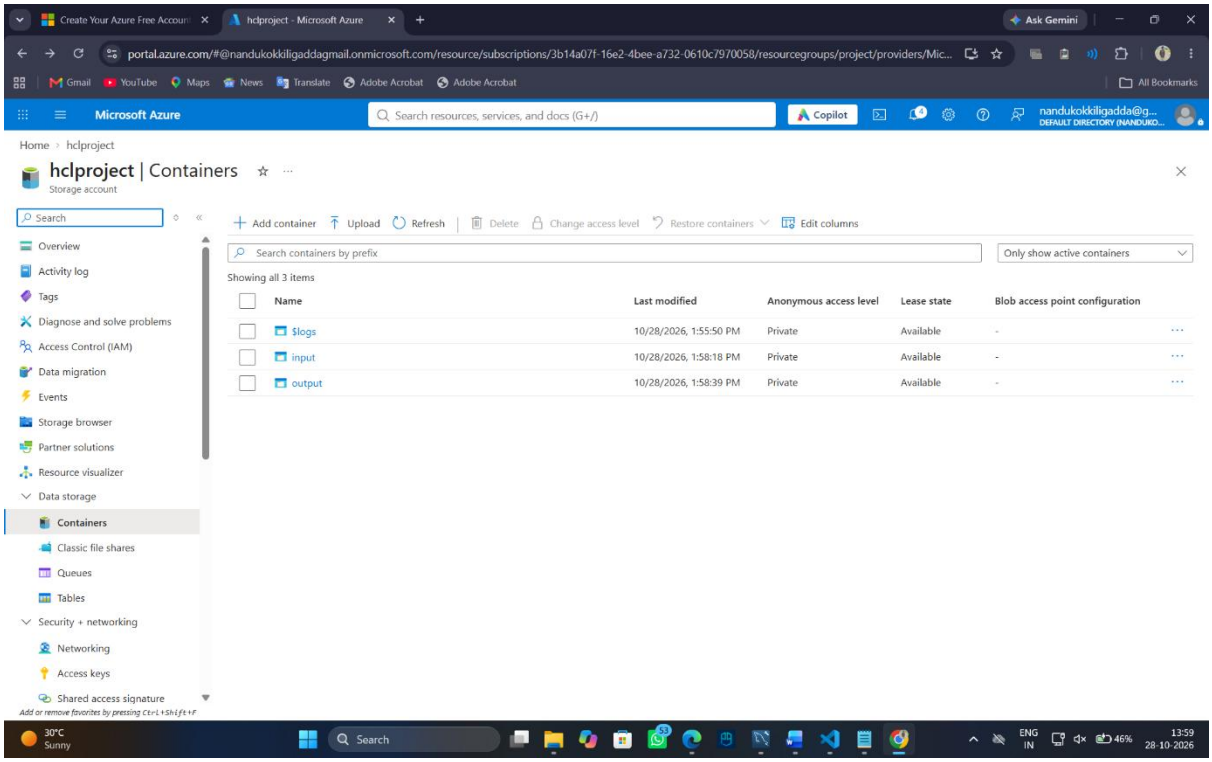

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Adobe Acrobat
All Bookmarks
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DEFAULT DIRECTORY (NANDUKO...
Home
Create a resource
Create a storage account
Basics
Advanced
Networking
Data protection
Security
Encryption
Tags
Review + create
Advanced
Azure Blob Storage
These are settings which apply to only Azure Blob storage
Enable hierarchical namespace
Enable network file system v3
Enable SFTP
Allow cross-tenant replication
Cross-tenant replication and hierarchical namespace cannot be enabled simultaneously.
Access tier
Hot
Optimized for frequently accessed data and everyday usage scenarios
Cool
Optimized for infrequently accessed data and backup scenarios
Cold
Optimized for rarely accessed data and backup scenarios
Azure Files
Next
Review and create
Give feedback
30°C
Sunny
Search
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28-10-2026

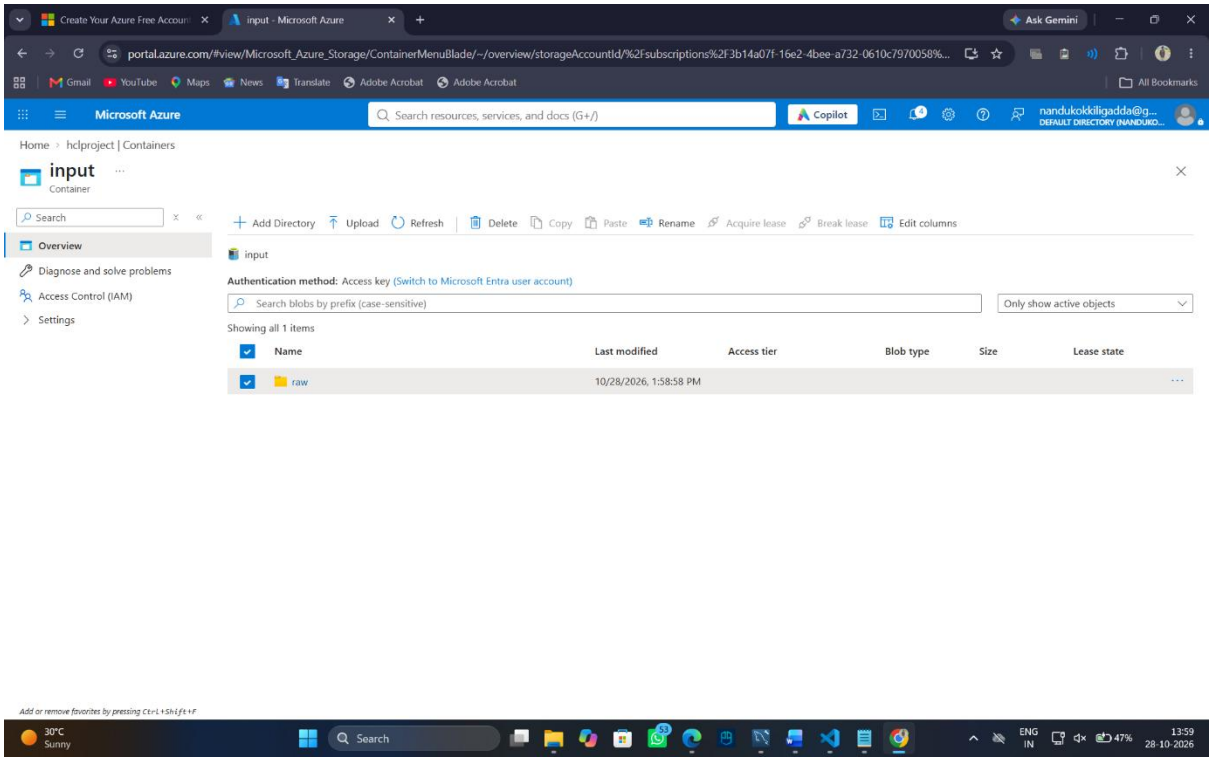


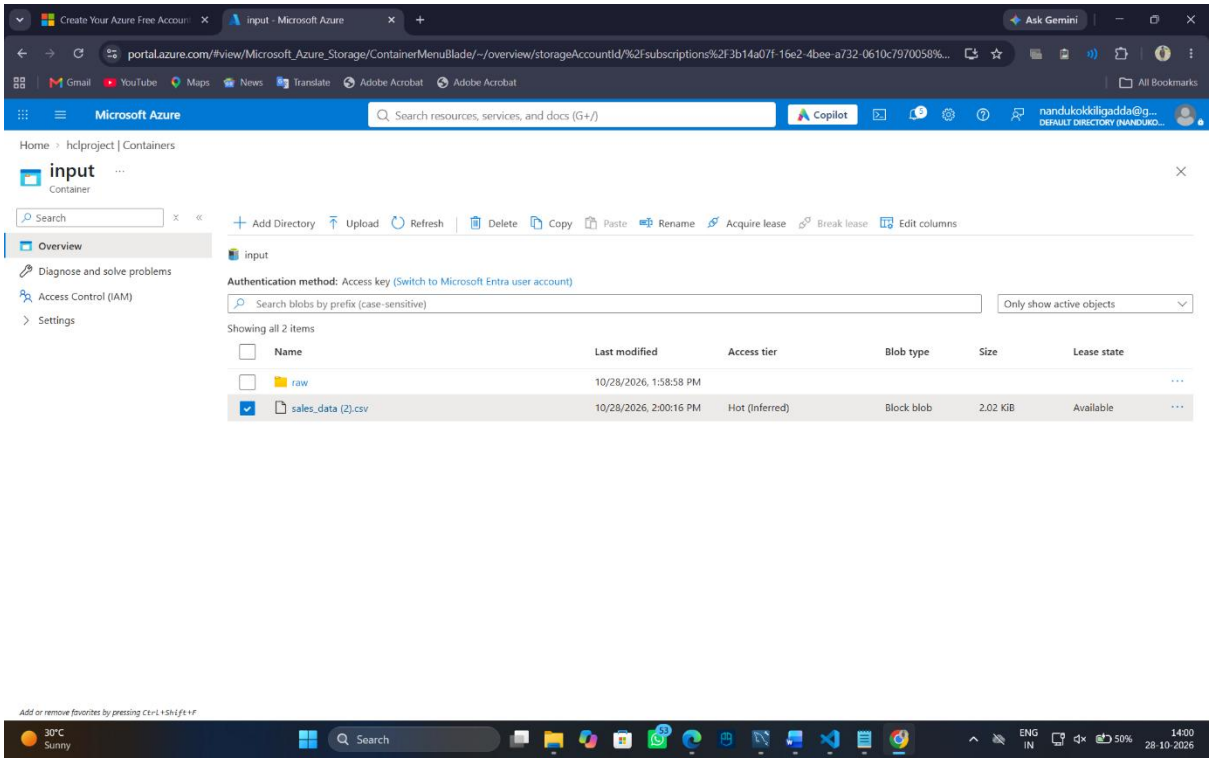
b. Create a filesystem or container and upload a sample dataset.




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Microsoft Azure
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Copilot
nandukokkiligadda@g...
DEFAULT DIRECTORY (NANDUKO...
Home > hclproject_1793175916753 | Overview > hclproject
hclproject | Containers
Storage account
Successfully created storage container
Successfully created storage container 'input'.
Search
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Partner solutions
Resource visualizer
Data storage
Containers
Classic file shares
Queues
Tables
Security + networking
Networking
Access keys
Shared access signature
Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Only show active containers
Showing all 2 items
Name
Last modified
Anonymous access level
Lease state
Blob access point configuration
$logs
10/28/2026, 1:55:50 PM
Private
Available
input
10/28/2026, 1:58:18 PM
Private
Available
30°C
Sunny
Search
ENG
IN
45%
13:58
28-10-2026


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portal.azure.com/#@nandukokkiligaddagmail.onmicrosoft.com/resource/subscriptions/3b14a07f-16e2-4bee-a732-0610c7970058/resourcegroups/project/providers/Mic...
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Home > hclproject
hclproject | Containers
Storage account
Search
Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Partner solutions
Resource visualizer
Data storage
Containers
Classic file shares
Queues
Tables
Security + networking
Networking
Access keys
Shared access signature
Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Only show active containers
Showing all 3 items
Name
Last modified
Anonymous access level
Lease state
Blob access point configuration
$logs
10/28/2026, 1:55:50 PM
Private
Available
input
10/28/2026, 1:58:18 PM
Private
Available
output
10/28/2026, 1:58:39 PM
Private
Available
30°C
Sunny
Search
ENG
IN
46%
13:59
28-10-2026





c. Explain how Microsoft Entra ID and Azure role-based access control can manage access to the data.

Microsoft Entra ID provides identity and authentication. Azure role-based access control (RBAC) assigns permissions to users, groups and service principals.

Suitable roles include:

Storage Blob Data Reader: Read data without modifying it.

Storage Blob Data Contributor: Read, write and delete data within the assigned scope.

d. Explain how you would verify that unauthorized users cannot access the dataset.

Assign the least-privilege role at the appropriate storage account, container or supported directory scope. Azure Data Lake Storage Gen2 also supports POSIX-style access control lists (ACLs) for directory and file permissions.

Question 16: Azure VM Troubleshooting

a. Verify the VM's provisioning and power states.

In the Azure Portal, navigate to Virtual Machines and select the VM. Check:

Provisioning state: Succeeded, Failed or another reported state.

Power state: Running, Stopped, Deallocated or another reported state.

b. Explain how Network Security Group rules can prevent connectivity.

NSGs contain inbound and outbound rules that permit or deny network traffic. A missing SSH rule for TCP port 22 or RDP rule for TCP port 3389 can prevent connections. Rules are evaluated according to priority.

c. Explain how subnet and routing configuration affect connectivity.

An incorrect subnet configuration, user-defined route, route table or next-hop setting can prevent traffic from reaching the VM. Verify the effective routes, subnet association, network interface configuration and any required gateway or peering connections.

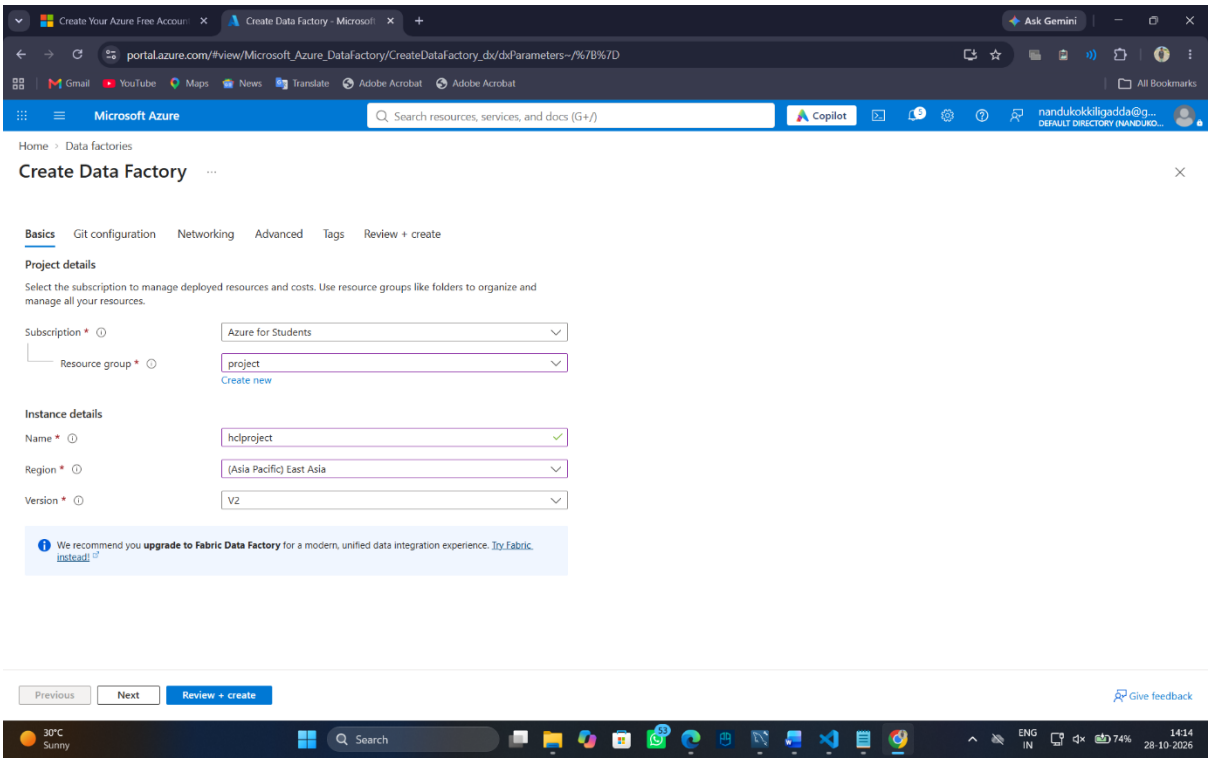
d. Recommend a secure troubleshooting method that avoids unnecessarily exposing the VM to the public internet.

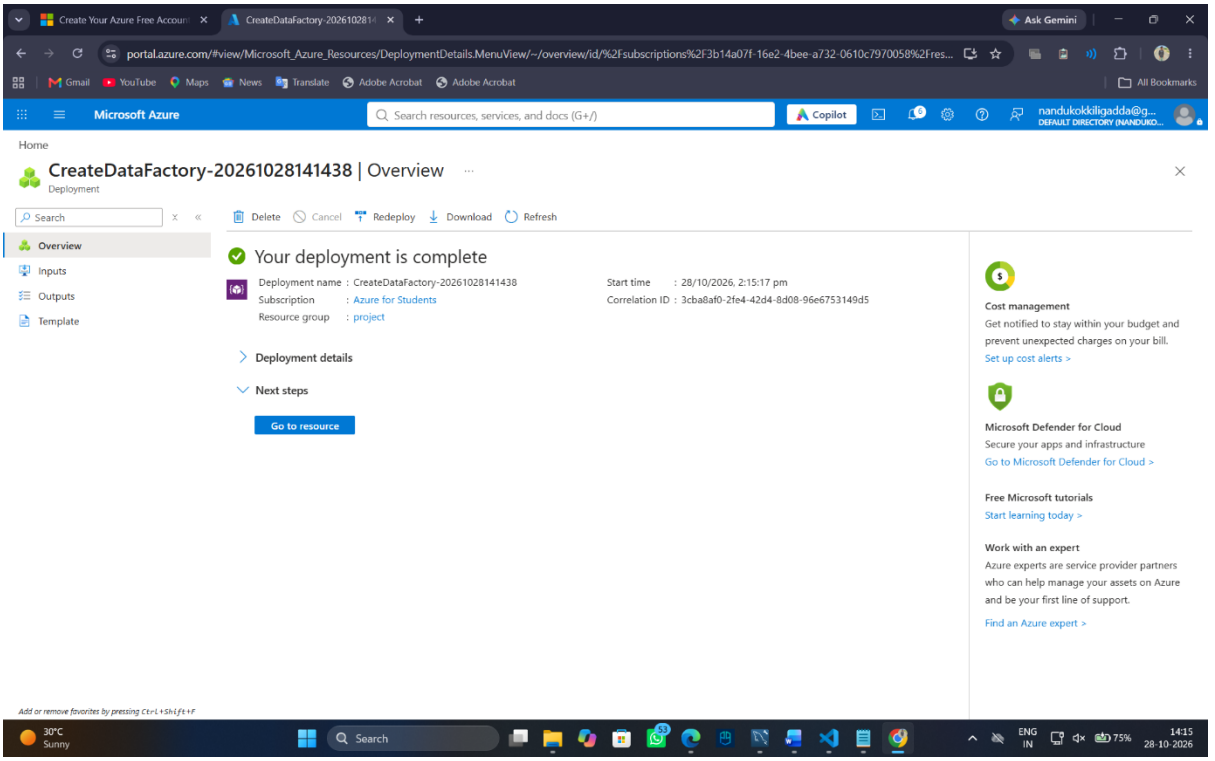
Use Azure Bastion to connect to the VM over SSH or RDP through the Azure Portal without assigning the VM a public IP address.
Additional checks:
Use Network Watcher connection troubleshooting.
Review NSG effective security rules and effective routes.
Check VM boot diagnostics and serial console where supported.
Avoid opening SSH or RDP to the entire internet.

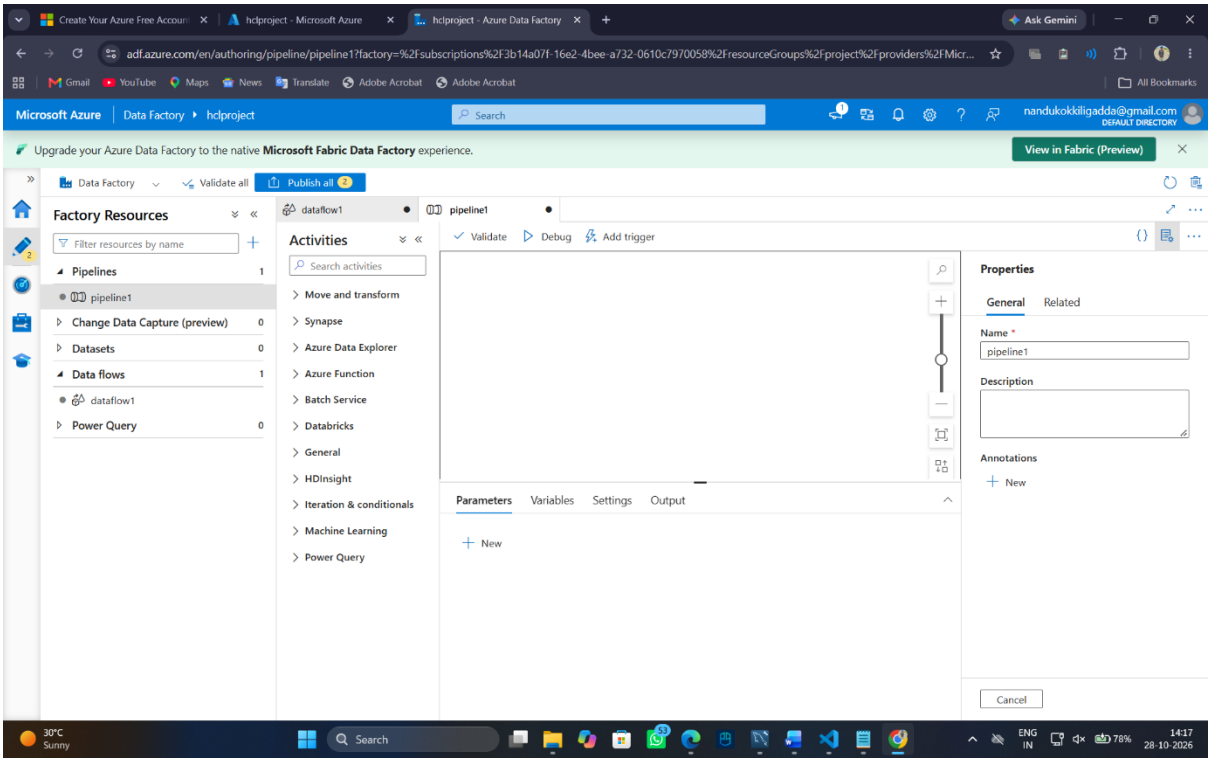
Question 17: Azure Data Integration

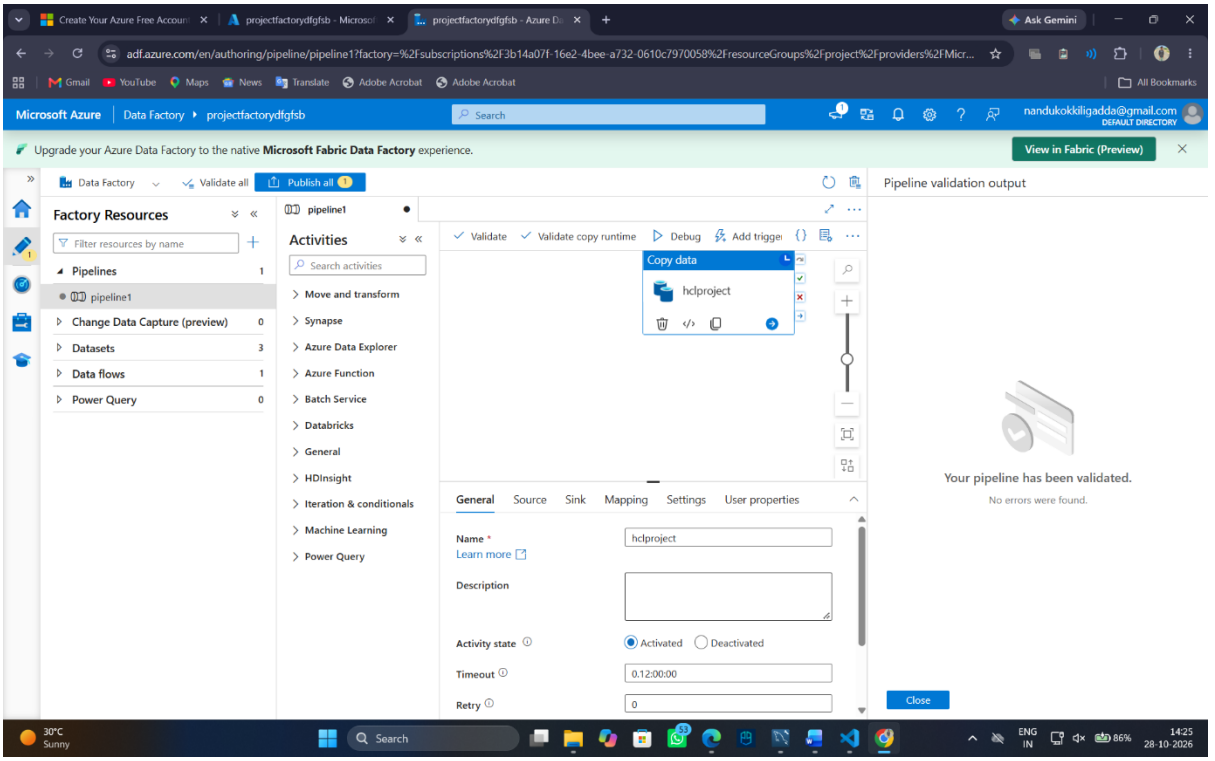
a. **Select an appropriate service to orchestrate data movement.**
   Use Azure Data Factory (ADF) to orchestrate and automate the data pipeline.

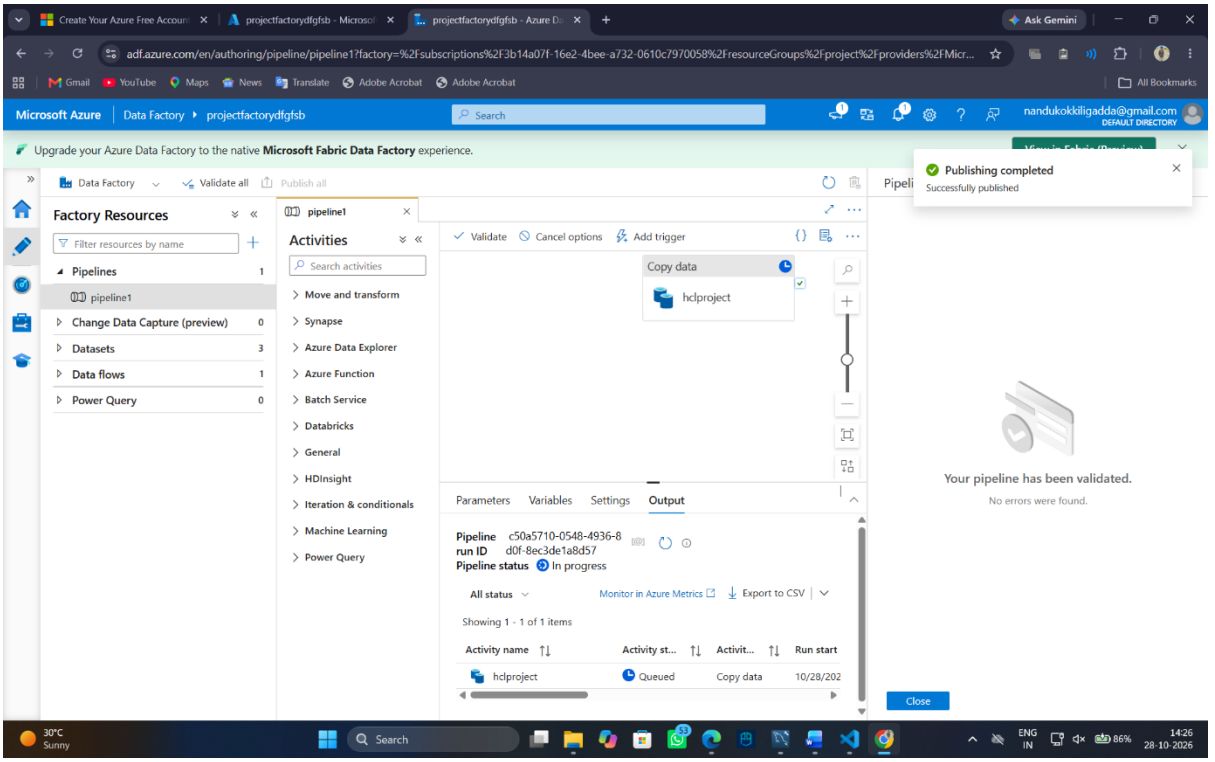
b. **Explain how to copy the CSV file into Azure Data Lake Storage Gen2.**

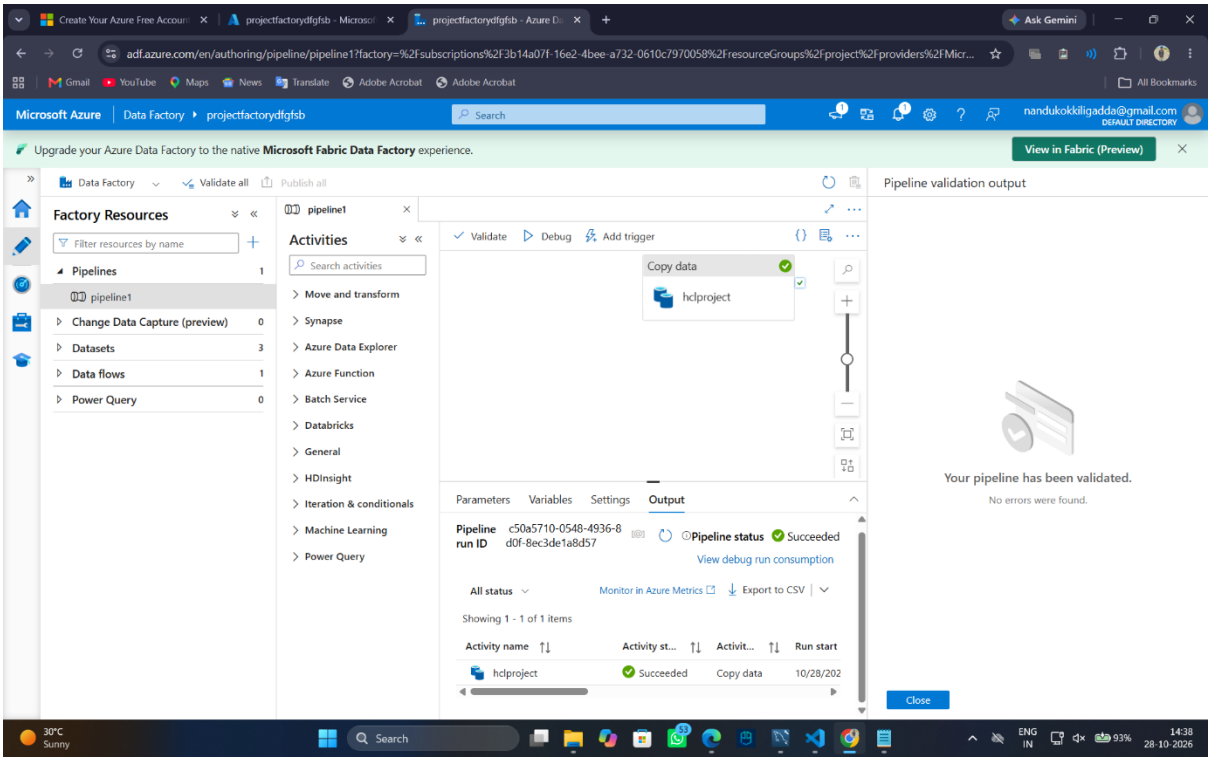



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Copilot
Home
CreateDataFactory-20261028141438 | Overview
Deployment
Delete
Cancel
Redeploy
Download
Refresh
Overview
Inputs
Outputs
Template
Your deployment is complete
Deployment name : CreateDataFactory-20261028141438
Subscription : Azure for Students
Resource group : project
Start time : 28/10/2026, 2:15:17 pm
Correlation ID : 3cba8af0-2fe4-42d4-8d08-96e6753149d5
Deployment details
Next steps
Go to resource
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
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Upgrade your Azure Data Factory to the native Microsoft Fabric Data Factory experience.
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Data Factory
Validate all
Publish all
Factory Resources
Filter resources by name
Pipelines
pipeline1
Change Data Capture (preview)
Datasets
Data flows
dataflow1
Power Query
Activities
Search activities
Move and transform
Synapse
Azure Data Explorer
Azure Function
Batch Service
Databricks
General
HDInsight
Iteration & conditionals
Machine Learning
Power Query
Validate
Debug
Add trigger
Parameters
Variables
Settings
Output
New
Properties
General
Related
Name *
pipeline1
Description
Annotations
New
Cancel


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Publish all
Factory Resources
Pipelines
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Change Data Capture (preview)
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HDInsight
Iteration & conditionals
Machine Learning
Power Query
Validate
Validate copy runtime
Debug
Copy data
hclproject
General
Source
Sink
Mapping
Settings
User properties
Name *
Learn more
Description
Activity state
Activated
Deactivated
Timeout
0.12:00:00
Retry
0
Pipeline validation output
Your pipeline has been validated.
No errors were found.
Close


Microsoft Azure
Data Factory
projectfactorydfgfsb
Upgrade your Azure Data Factory to the native Microsoft Fabric Data Factory experience.
Publishing completed
Successfully published
Factory Resources
Pipelines
pipeline1
Activities
Validate
Cancel options
Add trigger
Copy data
hclproject
Parameters
Variables
Settings
Output
Pipeline run ID
c50a5710-0548-4936-8d0f-8ec3de1a8d57
Pipeline status
In progress
All status
Monitor in Azure Metrics
Export to CSV
Showing 1 - 1 of 1 items
Activity name
Activity st...
Activit...
Run start
hclproject
Queued
Copy data
10/28/202
Your pipeline has been validated.
No errors were found.
Close



c. Describe how to organize files by ingestion date.

Use date-based directory partitioning,
for example:
/business-data/
year=2026/
month=10/
day=28/
.csv file

d.Explain how to handle malformed records and missing values.

Reject or redirect malformed records to a separate error folder. Apply appropriate missing-value rules, such as rejecting records with missing required IDs or retaining optional fields as null. Record error details for auditing and reprocessing.

e. Explain how to monitor pipeline execution and validate data completeness.

Use the ADF Monitor tab to inspect pipeline and activity status, duration, failures and error messages. Configure alerts for failures. Validate completeness by comparing source and destination record counts, checking required fields, and reconciling totals or checksums where appropriate.